
Credit Default Risk Prediction with Multi-Table Aggregation and Ensemble Boosting Models

Siqi Wang

swanggz@connect.ust.hk

Abstract

Financial inclusion remains a central goal in expanding access to credit and financial services, particularly when evaluating the repayment capacity of clients with limited or no credit history. The Home Credit dataset offers a comprehensive set of applicant features and auxiliary tables that capture historical credit behavior in rich detail. In this study, we present a strong, reproducible baseline for the Home Credit Default Risk prediction task by integrating domain-informed feature engineering with multi-table aggregations derived from credit and transactional records. Our approach adopts robust ratio features (e.g., payment rate, credit-to-income ratio), time-normalized indicators (e.g., age and tenure ratios), and systematic aggregations from the bureau, previous_application, installments_payments, credit_card_balance, and POS_CASH_balance tables. We employ logistic regression to ensemble LightGBM, CatBoost, and AdaBoost models, incorporating careful early stopping, iterative feature pruning, and seed ensembling. Finally, we release a modular pipeline designed to facilitate further research on fair and transparent credit scoring for advancing financial inclusion.¹

1 Introduction

Access to credit is fundamental to achieving financial inclusion, yet many individuals remain excluded due to limited or non-existent credit histories, leaving them vulnerable to exploitative lending practices. Home Credit seeks to mitigate this challenge by extending fair and accessible financial services to the unbanked, providing a safe and supportive borrowing experience. By leveraging alternative data sources, such as telecommunications, behavioral, and transactional information, Home Credit evaluates repayment capacity and broadens access to credit for underserved populations.

However, advancing financial inclusion requires developing robust models capable of accurately predicting repayment behavior, particularly for clients lacking formal credit histories. Credit default prediction has evolved from traditional statistical methods, such as logistic regression with handcrafted features, to modern machine learning approaches based on Gradient Boosting. Algorithms like LightGBM, CatBoost, and XGBoost have become standard tools for structured tabular data, offering strong predictive performance with relatively limited feature engineering. Many top-performing Kaggle solutions, including those in the Home Credit Default Risk competition, have demonstrated that boosting ensembles, when coupled with well-designed features, can substantially outperform conventional credit scoring models.

The Home Credit dataset provides a rich set of applicant characteristics and auxiliary tables that record historical credit behavior and financial interactions. In this study, we propose a pragmatic and reproducible framework for reliable default prediction, emphasizing both simplicity and interpretability. Our approach focuses on domain-informed feature engineering and boosting-based ensembling while deliberately pruning redundant or noisy features to maintain model transparency.

¹Github: <https://github.com/greenplumme/Home-Credit-Default-Risk-Project>

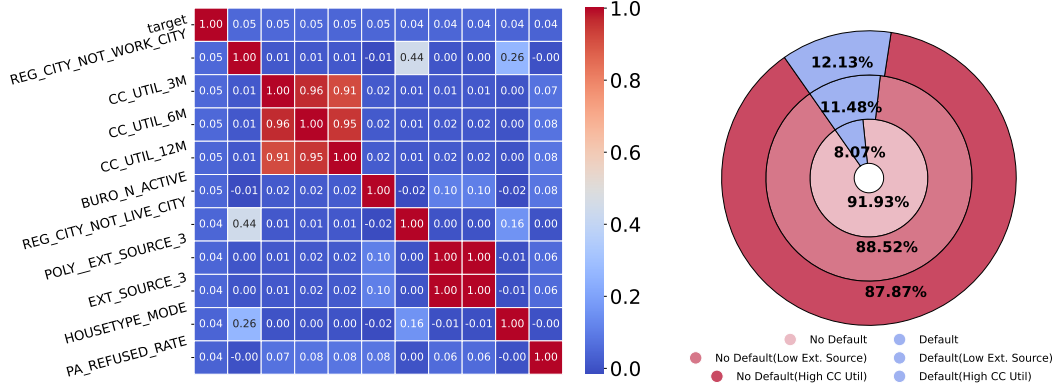


Figure 1: Correlation heatmap and feature–target distributions highlighting key drivers of default risk. (left) Correlation heatmap of selected high-impact features with the target variable, covering credit utilization, geographic mobility, and external source indicators. (right) Target distribution with respect to representative features (EXT_SOURCE_3 and CC_UTIL_12M); the inner circle shows the global target ratio, while outer layers illustrate class proportions across feature thresholds.

By employing a stacked ensemble combining LightGBM, CatBoost, and AdaBoost, and is further enhanced through stratified sampling, early stopping, and conservative regularization, we achieve competitive performance with a streamlined and reproducible pipeline. Overall, our method is designed to (i) perform robust feature engineering, (ii) integrate three high-performing models for ensemble prediction, (iii) deliver strong and reliable results, and (iv) provide systematic ablations to interpret model improvements.

Contributions. This work makes the following key contributions:

- A comprehensive feature engineering pipeline that integrates and aggregates data from six auxiliary tables and handles missing, categorical, and continuous features.
- An extensive exploratory data analysis (EDA) providing insights into key predictors.
- A robust ensemble model training pipeline leveraging stratified splits, early stopping, and regularization for stability.
- A detailed ablation and comparison study identifying the most influential features and optimal model configurations.

2 Exploratory Data Analysis Results

We use the official Home Credit Default Risk dataset, which includes a binary target variable, TARGET (1 = default). The main application tables (application_train/test) are augmented with several auxiliary historical tables that provide valuable insights into the applicants’ credit history.

2.1 Dataset Overview

The Home Credit dataset comprises a primary application table and six auxiliary tables capturing clients’ historical credit behavior.

1. **Application Train/Test:** Main applicant-level data (demographics, income, loan amount). The training file includes the target variable (TARGET).
2. **Bureau & Bureau Balance:** Historical credits from external credit bureaus, with monthly balance records per credit.
3. **Previous Application:** Records of all prior loan applications to Home Credit, including statuses and amounts.
4. **Installments Payments:** Detailed repayment history for the previously disbursed credits, including payment amounts and delays.

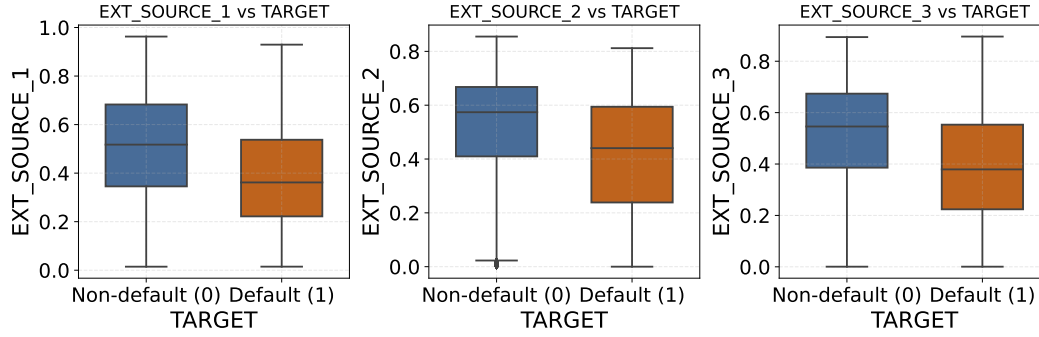


Figure 2: All three EXT_SOURCE features show clear separation across TARGET classes: borrowers with higher external source scores have lower default rates, consistent with their interpretation as creditworthiness proxies.

5. **POS-CASH Balance:** Monthly POS and cash loan performance data.
6. **Credit Card Balance:** Monthly records for previous Home Credit credit cards.
7. **Columns Description:** Metadata providing definitions for all variables.

2.2 Missing Patterns, Target Distribution and Imbalance

These tables enable multi-level modeling of a client's financial history to support a comprehensive credit risk assessment. However, In the main application table, over 50% of features show missing rates higher than half, particularly those related to housing and living conditions. The missing patterns demonstrate the need for a robust imputation strategy or the use of feature engineering to handle missing values efficiently.

In the training dataset, only about 8% of clients defaulted (TARGET = 1), while approximately 92% did not (TARGET = 0). This indicates a highly imbalanced dataset (shown in Figure 1 (right)). Accuracy alone is not a sufficient evaluation metric; metrics such as AUC, Recall, and Precision are more appropriate for a highly imbalanced dataset.

2.3 Categorical Feature Insights

Several categorical variables show clear differences in default rates.

1. CODE_GENDER: More female applicants, yet males tend to have higher default rates.
2. REGION_RATING_CLIENT_W_CITY: most applicants belong to region rating 2, but those from rating 3 regions have the highest default proportion.
3. NAME_EDUCATION_TYPE: most clients have "Secondary / Secondary special" education; the default rate is highest for "Lower secondary" and lowest for "Academic degree".
4. OCCUPATION_TYPE: laborers, drivers, and low-skill workers show higher default rates than managerial or professional occupations.
5. REG_CITY_NOT_WORK_CITY: clients working outside registered city are more likely to default.
6. ORGANIZATION_TYPE: industries like transport, construction, and restaurants exhibit higher default risk, whereas police, universities, and government institutions show lower rates.
7. NAME_CONTRACT_STATUS (from previous applications): clients with past refusals or canceled applications tend to have higher default probabilities.

2.4 Continuous Feature Insights

1. **Correlation Analysis:** Using Phi-K and Pearson correlation analyses, most feature pairs show weak linear relationships, as reflected by the light-colored heatmap in Figure 1 (left).

Exceptions include housing-related variables (e.g., area, floor), which exhibit strong mutual correlations, and significant linear relationships between `AMT_GOODS_PRICE`–`AMT_CREDIT` and `DAYS_EMPLOYED`–`DAYS_BIRTH`.

2. **Key Predictive Features and Distribution Patterns:** Among all continuous features, the `EXT_SOURCE_1/2/3` group demonstrates the highest correlation with the target variable, making them key predictive signals (shown in Figure 2). Defaulters generally exhibit lower values for the `EXT_SOURCE` series, while non-defaulters concentrate in higher-value regions. Similarly, defaulters tend to be younger (lower median and quartile ages, as indicated by `DAYS_BIRTH`), have shorter credit histories (`DAYS_CREDIT`), and repay fewer installments (`CNT_INSTALLMENT_MATURE_CUM`). Non-defaulters typically show the opposite trends.
3. **Data Quality:** Several `DAYS_*` variables contain sentinel values (e.g., 365243), which should be replaced with NaN to avoid distorting distributions.

2.5 Summary of EDA Conclusions

1. The main and auxiliary tables should be merged using the client ID, with careful handling of missing values to retain useful information.
2. Certain categorical features (e.g., occupation, city mismatch, past refusals) exhibit strong discriminatory power and should be included in feature engineering.
3. The `EXT_SOURCE` group provides the strongest correlation with the target and is critical for feature interaction design.
4. Sentinel values (e.g., in `DAYS_*`) must be cleaned, as they distort the feature distributions.
5. Most features are weakly correlated, which reduces redundancy; however, some highly collinear groups require monitoring.
6. Given the extreme class imbalance, subsequent modeling should account for weighting, sampling, or metric adjustments.

3 Method

3.1 Data Preprocessing

Building on the EDA insights, the preprocessing steps followed were as follows:

1. **Missing Value Imputation:** Missing values were filled using the median values computed from the training set, ensuring that test set values were treated consistently.
2. **Categorical Data Encoding:** Non-numerical categorical variables were transformed into numeric representations using mapping techniques.
3. **Feature Scaling:** Numerical features were standardized or scaled where necessary to ensure they had similar ranges and distributions. This helps in improving the performance of the model training and prediction.
4. **Polynomial Feature Generation:** Polynomial Features were applied to generate interaction terms and non-linear relationships among features, providing more expressiveness in the model training and prediction.

We performed aggregation on the auxiliary tables without causing row explosion. Specifically, we aggregated features from the following sources:

1. `bureau` and `bureau_balance`: External credit history and monthly credit status.
2. `previous_application`: Previous loan applications (approved/refused).
3. `installments_payments`: Installment schedules and realized payments.
4. `credit_card_balance`: Revolving credit behavior.
5. `POS_CASH_balance`: Point of sale/loan performance over time.

The final features were used to train models with stratified splits, ensuring that the training and validation datasets had balanced distributions of the target variable. We report the performance of our models using AUC on an 80/20 stratified holdout split.

Table 1: Key LightGBM hyperparameters used in model training. Hyperparameters were tuned using grid and random search to balance performance and regularization.

Parameter	Value / Range	Description
n_estimators	5000 (early stop)	Max boosting rounds
learning_rate	0.02–0.03	Step size per iteration
max_depth	10–15	Limits tree depth
num_leaves	30–64	Controls model complexity
feature_fraction	0.7	Fraction of features per tree
bagging_fraction	0.75	Fraction of samples per iteration
bagging_freq	3	Bagging frequency
max_bin	400	Discretization granularity
reg_lambda	5.0	L2 regularization
min_child_samples	120	Minimum samples per leaf

3.2 Model and Training Settings

We employ LightGBM, CatBoost, and AdaBoost, which are ensembled using logistic regression as the meta-learner. These gradient boosting frameworks are highly efficient for large-scale tabular data and well suited for binary classification tasks such as credit default prediction. The stacked model was trained using an 80/20 stratified train–valid split, with early stopping applied to prevent overfitting and to determine the optimal number of boosting iterations. The training objective is binary classification, predicting whether an applicant will default on a loan.

Model performance was optimized with respect to the Area Under the Receiver Operating Characteristic Curve (AUC), a robust and widely used evaluation metric for imbalanced datasets. To assess generalization, we performed cross-validation, and employed iterative feature pruning to eliminate weak predictors and improve interpretability. The pruning process was guided by gain-based feature importance, as illustrated in Figure 4.

To further improve stability, we averaged predictions from three different random seeds (seed ensembling). Hyperparameters of typical model LightGBM are summarized in Table 1, which were determined through a combination of grid search and random search for balanced accuracy and regularization strength (details provided in the code repository).

3.3 Pipeline Overview

In this section, we outline the key steps in constructing the feature engineering and model training pipeline for the Home Credit Default Risk task. The pipeline integrates multiple data sources and applies domain-specific feature engineering to optimize training and prediction performance.

```
Input: application_train/test, bureau, bureau_balance,
       previous_application, installments_payments, credit_card_balance,
       POS_CASH_balance
```

- 1: Build X_{app} = core features + domain-driven ratios + time-normalized features + external source interactions
- 2: For each table T:
 - feats_T = groupby(SK_ID_CURR).aggregate(domain_stats)
 - if time index: add recent-window stats (3/6/12m)
- 3: Merge all feats_T into X_{app} on SK_ID_CURR
- 4: Add degree-2 interactions among EXT_SOURCES and age
- 5: Split stratified -> (X_{train} , X_{valid})
- 6: Train LightGBM, CatBoost, XGBoost with early stopping
- 7: Stack these three models using logistic regression
- 8: Drop low-gain features, retrain (optional)
- 9: Average multi-seed predictions for stability

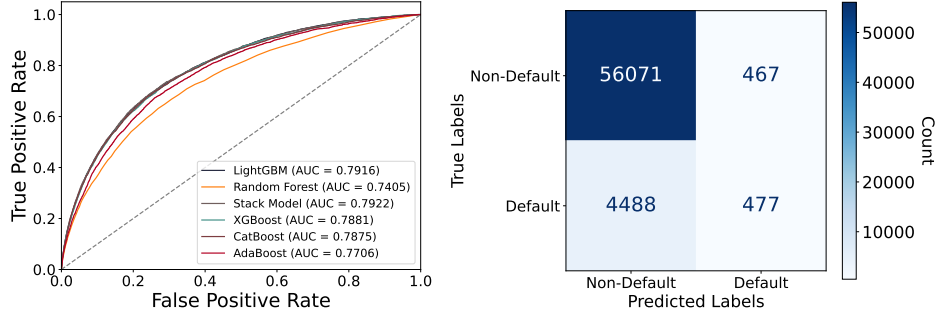


Figure 3: Confusion matrix and ROC curve of the different models on the validation set. The ROC curve (left) illustrates discrimination capability compared with baseline models, while the confusion matrix (right) summarizes prediction outcomes of the stack model.

4 Experiments

In our experiments, we combined table aggregation, domain features and polynomial terms and conduct ablation experiments to evaluate and analyze. Furthermore, we conduct comparison experiments across different commonly used models, and evaluate their performance across different metrics.

4.1 Experimental Details

Our pipeline builds an ensemble method by combining LightGBM, XGBoost, and CatBoost models. The dataset is first split into training and validation sets using stratified sampling to preserve class balance. Each model is trained with early stopping and optimized hyperparameters to capture different learning patterns. Their validation predictions are then blended using weighted averaging or a meta-model (such as logistic regression) in a stacking approach, which learns how to best combine their outputs. Finally, the ensemble’s predicted probabilities are applied to predict.

We evaluated its performance on the validation set using AUC as the metric. The AUC score achieved on the validation set was 0.792, indicating strong predictive ability. The model performed well both on the public and private leaderboards, with a slight improvement on the private leaderboard, suggesting that the model generalizes well on unseen data.

4.1.1 Ablation Experiments

We conduct ablation experiments on LightGBM. Table 2 presents the results of the ablation study, where we assess the impact of different feature sets on model performance. Firstly, we use App-Train Features Only. The model trained solely on the app-train features yields a solid baseline, achieving an AUC of 0.773 and a recall of 0.037. However, the model struggles with detecting defaults, as evidenced by the low recall and F1-score.

Then we include polynomial terms of EXT_SOURCE and age features. However, it brings only slight gains in precision (0.539) and a marginal increase in accuracy (0.920). Furthermore, we aggregate features from auxiliary tables (e.g., bureau, installments, and previous applications), which leads to an improvement in AUC (0.791) and recall (0.061). This confirms that data aggregation across behavior tables introduces valuable signals, improving the model’s ability to predict defaults, especially for hard-to-classify cases.

Overall, the combination of both polynomial terms and all-table aggregation delivers the highest AUC (0.792), Recall (0.061) and F1-score (0.110), alongside a relatively good precision. Importantly, this combination maintains the highest accuracy of 0.920, reinforcing the importance of leveraging both feature engineering techniques for capturing non-linear relationships and utilizing multi-table aggregations for richer, more diverse data signals.

Summary. The ablation study demonstrates that while polynomial terms alone do not significantly improve model performance, their combination with all-table aggregation results in the

best-performing model. This indicates that multi-table aggregation, along with targeted feature transformations, is crucial for enhancing the model’s predictive power in imbalanced settings like the Home Credit Default Risk task. Compared with the App-Train only baseline (AUC = 0.773), our final pipeline achieves 2.4 % AUC gain and 62 % improvement in F1-score.

Table 2: Ablation study comparing different feature sets on the app_train dataset. The top-performing model in each metric is highlighted in bold.

Feature Set	AUC	Precision	Recall	F1-Score	Accuracy
App-Train Features Only	0.773	0.529	0.037	0.068	0.919
+ Polynomial Terms	0.773	0.539	0.035	0.066	0.920
+ All-Table Aggregation	0.791	0.516	0.061	0.109	0.919
+ Poly + Aggregation	0.792	0.518	0.061	0.110	0.920

4.1.2 Comparison Experiments

We have included the models commonly used in repayment prediction for comparing, such as Logistic Regression, Random Forest, XGBoost, etc. For the evaluation metrics, we adopt AUC (Area Under the ROC Curve) as a key metric for imbalanced classification problems. Besides, we used Precision, the proportion of true positives out of all predicted positives; recall, proportion of true positives out of all actual positives; F1-Score, Harmonic mean of precision and recall; Accuracy, Overall accuracy of the model in predicting correct labels.

In Table 3, we compare the performance of these models on the validation set. The results clearly show the progression from linear to ensemble methods and the benefits of incorporating domain-specific features. Specifically, Logistic Regression provides a weak baseline. Its linear decision boundary fails to capture the complex nonlinear relationships between applicant attributes and default risk. While Random Forest slightly improves AUC to 0.7405 but still struggles with recall, indicating limited adaptability to the heavy class imbalance. AdaBoost, CatBoost, AdaBoost and XGBoost show stronger discriminative power, with LightGBM achieving substantially higher AUC and Recall, reflecting its ability to model hard-to-classify minority cases. AdaBoost was evaluated but excluded from the final stack due to lower recall. Our stacked model outperforms all baselines, achieving the highest AUC (0.7922) and recall (0.0961), confirming that combining multiple strong learners with leakage-safe multi-table aggregation and domain-driven feature ratios yields superior predictive power beyond what single models or raw boosting can achieve.

Summary. Overall, ensemble gradient-boosting methods demonstrate the best generalization performance under strong regularization. The integration of domain-specific ratios, polynomial interactions, and table-level aggregations substantially enhances model robustness while preserving interpretability. These findings validate that thoughtful feature engineering, coupled with structured ensemble learning, consistently outperforms generic baselines in credit default prediction.

Table 3: Model performance comparison on validation data. The top-performing model in each metric is highlighted in bold.

Model	AUC	Precision	Recall	F1-Score	Accuracy
Logistic Regression	0.5747	0.1250	0.0004	0.0008	0.9191
Random Forest	0.7405	0.7037	0.0038	0.0076	0.9195
CatBoost	0.7875	0.6204	0.0342	0.0649	0.9203
LightGBM	0.7916	0.5178	0.0614	0.1098	0.9196
AdaBoost	0.7706	0.5152	0.0274	0.0520	0.9194
XGBoost	0.7881	0.5789	0.0421	0.0785	0.9202
LR-Stack (LightGBM + CatBoost + XGBoost)	0.7922	0.5053	0.0961	0.1614	0.9194

4.2 Feature Importance

Feature importance analysis highlights the demographic and behavioral signals underlying default prediction. As shown in Figure 4 (left), the model’s predictive power is largely driven by

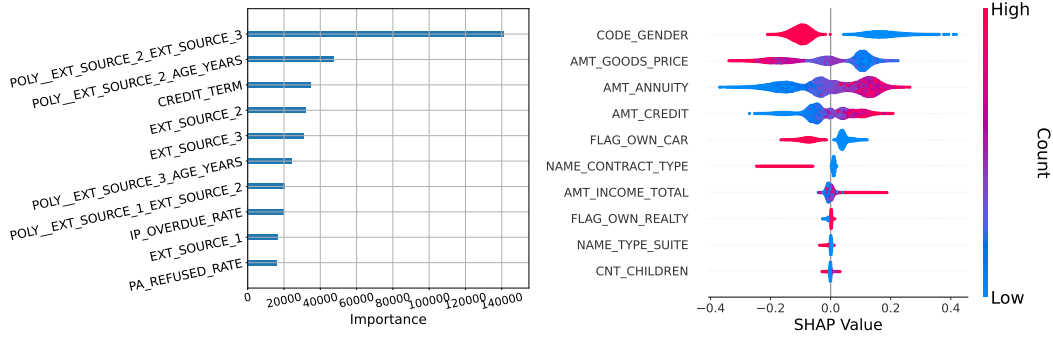


Figure 4: Top features contributing to default prediction. (Left) LightGBM gain-based feature importance highlights credit utilization, and external source scores as dominant predictors. (Right) SHAP value distributions confirm consistent feature effects aligned with domain intuition.

externally derived credit scores and interaction-based features. The most influential predictors include combinations of external sources such as POLY_EXT_SOURCE_2_EXT_SOURCE_3, EXT_SOURCE_3, and EXT_SOURCE_2, as well as key financial ratios like CREDIT_TERM and AMT_ANNUITY. Interaction terms between external credit scores and applicant age further enhance predictive strength, indicating that nonlinear relationships between external risk indicators and demographic variables are crucial for default prediction. In contrast, the least important features include various FLAG_DOCUMENT indicators, categorical administrative flags, and rarely activated credit bureau variables, suggesting they contain limited discriminatory information for assessment.

In Figure 4 (right), SHAP value analysis confirms these findings and provides new insights at the instance level. Financial attributes such as AMT_CREDIT, AMT_ANNUITY, and AMT_GOODS_PRICE exhibit positive SHAP contributions. Larger credit amounts increase default likelihood, whereas higher income (AMT_INCOME_TOTAL) and smaller household size (CNT_CHILDREN) are associated with lower risk. Demographic variables (CODE_GENDER, FLAG_OWN_CAR, FLAG_OWN_REALTY) also modulate repayment likelihood. Overall, the SHAP confirms that behavioral stability, financial exposure, and demographic context are the principal drivers of credit default prediction.

4.3 Discussion

Our findings reveal several key insights: (1) domain-specific ratios (e.g., payment-to-income, credit-to-annuity) and time-normalized features (e.g., age and credit tenure ratios) substantially enhance model interpretability and predictive reliability; (2) leakage-safe aggregation across behavioral tables is the primary source of performance gain; (3) polynomial interactions between key features provide modest yet consistent improvements; (4) ensemble models outperform single learners in both accuracy and generalization; and (5) lightweight feature pruning and seed averaging offer a cost-effective means to improve model stability.

For error analysis, we examine score distributions across age groups, income deciles, and employment tenure. Common failure cases involve applicants with sparse credit histories, inconsistent EXT_SOURCE signals, or near-zero income and credit values.

5 Conclusion

This study introduces a reproducible and interpretable framework for credit default prediction using the Home Credit dataset. We began with a thorough exploratory data analysis to uncover key relationships between features and target, which informed the design of preprocessing pipeline including categorical encoding, feature scaling, imputation, and polynomial interaction construction. Building on these, we developed a compact stacked ensemble model combining LightGBM, CatBoost, and AdaBoost. Comprehensive ablation and comparative experiments demonstrate that domain-informed features, leakage-safe multi-table aggregations, and regularized boosting techniques are crucial for achieving both accuracy and generalization. Overall, the proposed pipeline provides a transparent and efficient foundation for future research on fair and inclusive credit scoring.